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Section 2
Cloud-Based ERP vs On-Premise ERP: Future Directions for Enterprises
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Executive Summary

This report analyses the trends of Enterprise Resource Planning (ERP) deployment models by comparing cloud-based and on-premise systems to understand future directions for enterprises. ERP integrates all major business functions such as finance, accounting, customer service, operation and marketing into a unified system. Traditional on-premise ERP provides more control and user customisation while cloud-based ERP allows remote work, is more flexible and scalable.

On-premise ERP and Cloud-based ERP is compared across 8 key areas, that is setup, cost, access, maintenance, customisation, security, best targeted area and future trends. The finding shows that cloud ERP is favoured by small and medium-sized organisations for its flexibility, low upfront cost and constant feature development. On the other hand, on-premise ERP is preferable by large enterprises that need total control over data and tailored functions.

The study concludes that hybrid ERP which combines both cloud and on-premise systems will shape the future of enterprise systems. It is recommended that organisations consider total cost of ownership, regulations and long-term scalability before choosing a system and include technologies such as analytics and artificial intelligence to support better decision making.

1.0 Introduction

Enterprise Resource Planning (ERP) systems is widely used to manage medium to large scale business, as it enables integration of almost every core business functionality and data accountability across finance, supply chain, human resources and customer management. Traditionally, ERP systems were deployed on-premise, local-hosted within a company's own infrastructure and managed by in-house IT departments. On the contrary, Cloud-based ERP is hosted remotely by service providers and users access it via the Internet. Cloud-based ERP has emerged as the dominant deployment model for medium to large-scale enterprises. According to recent global analytics, cloud-based ERP shows dominance and signaling a future direction increasingly defined by flexibility, scalability, and service-based architectures (Grand View Research, 2024).

This report aims to define the future trend of ERP deployment, whether enterprises will continue migrating toward cloud-based systems or retain on-premise infrastructure for strategic or regulatory reasons. The analysis considers both large enterprises and small to medium-sized organisations where ERP systems play an important role in improving efficiency and supporting decision making across different business levels. Large organisations with complex operations may still rely on on-premise or hybrid ERP solutions due to the greater ability to control over data, compliance and integration needs. In contrast, smaller and growing businesses often choose

cloud-based ERP for its lower cost, flexibility and easier access. The study evaluates both deployment models across several areas including cost, maintenance, security, scalability and customisation to determine their suitability for different types of organisations.

This study focused on the structured comparison between cloud-based and on-premise ERP systems in eight core areas: setup, cost, access, maintainability, customisability, security, best targeted area and future trends. This report aims to provide a clear understanding of the enterprise priorities and alignment with each model. In such, insights of the strategic consideration that drives ERP deployment decisions in large organisations will be gained.

2.0 Literature Review

Enterprise Resource Planning (ERP) systems is the integrated software that combines core business functions such as finance, supply chain, human resources and customer management in a single database. This makes the decision of cloud-based or on-premise deployment models a key determinant of system performance, flexibility and cost efficiency (Elmonem et al., 2017, 1-2). For decades, on-premise ERP systems were favoured for their ability to have customisation and controls over the system. However, the rise of cloud computing offers a new subscription based option accessible with the Internet (Lee et al., 2024). Hybrid and two-tier ERP strategies where cloud systems support subsidiaries while on-premise systems remain at headquarters are increasingly seen as transitional models for large enterprises (Johansson et al., 2025).

Prior comparative studies have stated that cloud-based ERP provides faster implementation, lower upfront capital cost and improved accessibility, while on-premise ERP are more preferable by organisations that require lots of customisation and strict data security (Salih et al., 2021). There are several literature reviews that highlight cloud ERP on their operational agility and support of global collaborations, but challenges remain in data security, integration complexity and vendor lock-in (Elmonem et al., 2017, 5). Earlier available research focused on cost and adaptation factors, more recent works are studying the long-term value generation, interoperability and sustainability outcomes.

Adaptation issues that are present in the literature include organizational size, IT maturity, industry regulation and strategic priorities (Salih et al., 2021). Emerging technical trends such as microservices-based architectures, API-first integration and Integration Platform-as-a-Service (iPaaS) enhances ERP interactions with external ecosystems and cloud infrastructures that improves modularity and scalability. However, this causes the emergence of new challenges of management and system interconnection when data is coordinated, shared and managed with the cloud vendors.

Security and compliance continue to be major discussion points in the literature. Although cloud providers spend heavily on security standards and system reliability, many organisations are still

worried about where their data is stored and losing direct control over it. There are still research gaps in long-term cost comparisons, business performance after migration and real-world evidence from small and medium enterprises in developing regions. Filling these gaps is important for understanding how ERP systems are changing and for helping organizations choose the best deployment models for the future.

3.0 Analysis and Discussion

Enterprise Resource Planning (ERP) systems support companies manage finances, inventory and human resources. On-premise ERP and cloud-based ERP are the two major types. Cloud-based ERP is vendor-managed and operates online. It updates automatically, is easy to use from anywhere, and typically has lower starting costs. For example, A Malaysian SME adopted Oracle NetSuite to streamline operations and reduce IT overhead. (Netsense, 2025) Because of this, small and medium-sized organisations who desire flexibility and quick setup will find it suitable.

On-premise ERP is operated by internal IT staff and placed on company servers. Although it costs more in the beginning and requires ongoing maintenance, it provides users more control and customisation. This style is frequently used by large organisations or those with strict data rules. As an example, a manufacturing firm retained SAP ECC for deep customization and data control. (Hale, 2018)

Cloud ERP is becoming increasingly popular since it allows for remote work and can grow with the organisation. However, for security and control, some companies prefer to choose on-premise ERP. Many companies may employ a hybrid ERP in the future, combining the two and including smart features like analytics and artificial intelligence(AI) to enhance decision-making. (Cloud ERP vs. On-Premise ERP: A Comprehensive Business Guide, 2025)

Feature	Cloud-Based ERP	On-Premise ERP
Setup	Online,managed by vendor	Installed on company servers
Cost	Monthly subscription,cheaper upfront	Expensive setup, long-term ownership
Access	Anywhere with internet	Only within company network
Updates	Automatic by vendor	Manual, done by company IT
Customization	Limited	High
Security	Vendor handles it, may raise concerns	Full control, but needs strong IT team
Best For	Small to medium businesses	Large companies

Future Trend	Hybrid ERP, AI and smart features	Slowly moving toward cloud or hybrid
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Table 3.1 Cloud-Based vs On-Premise ERP

4.0 Recommendations

Companies should carefully consider their long-term goals, their budget and any regulations they may need to follow before selecting an ERP system, especially if they work in fields like healthcare or finance. Cloud-based ERP can often be a better option for small and medium-sized companies or those looking to grow rapidly because it is simpler to set up, less costly up front and accessible from anywhere. However, on-premise ERP or a combination of both (known as hybrid ERP) may be better appropriate for companies that want highly specialised capabilities or have strict control over their data.

Furthermore, companies should first evaluate the five-year total costs of each option, including the initial cost as well as maintenance, upgrades and support, in order to make a smart decision. They should also look at the ERP system's ability to expand with the business, its simplicity of integration with other tools and the vendor's availability for support. A hybrid ERP could help a company lower risks and adopt cloud computing gradually if it is not sure it wants to make the whole transition. Finally, it is important to select an ERP system with intelligent features like artificial intelligence (AI) and data analytics, which could help organizations make smarter decisions faster in the future.

5.0 Conclusion

The decision between cloud-based and on-premises ERP systems shows how companies are changing their risk management, technology use and operations. Due to its flexibility, low starting costs and ongoing feature development, cloud ERP is becoming more and more popular. However, on-premise ERP might be helpful for companies that need total control or specific customisations. Many companies are going to add intelligent technologies like artificial intelligence (AI) and combine the two (hybrid ERP) to help them run more effectively. Companies should think about their long-term objectives and how technology could support their growth in order to make the optimal choice.

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